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AI Powered Developer Platforms and Tools Experiential Learning

COLLEGE ADMISSIONS HELPDESK BOT

Gemini-based Admissions Assistant Chatbot

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1. Executive Summary

The College Admissions Helpdesk Bot is a prototype conversational assistant designed to answer prospective students' admissions and student-policy questions for the fictional Sahyadri School of Business. The prototype uses a Gemini Gem with an uploaded Excel Knowledge Base as its source of truth. The project focuses not only on answering routine FAQs, but also on handling ambiguity, avoiding unsupported claims, recognizing out-of-scope requests, and referring appropriate cases to a human admissions team.

The project documentation defines a clear operating model: the bot should answer only from the Knowledge Base, ask a short clarification question when a query is ambiguous, avoid assumptions, and hand off requests when information is unavailable, an exception or individualized decision is requested, a complaint or dispute is raised, private information is requested, clarification does not resolve the issue, or the request is outside the helpdesk scope.

The completed hand-off evaluation contains 5 test cases. The bot made the expected hand-off decision in 4 cases, giving a hand-off decision accuracy of 80%. It actually handed off 3 of the 5 cases, giving a hand-off rate of 60%. Four cases were handled correctly; the only failed case was a refund complaint/dispute, where the expected action was human review but the recorded actual action was "Do not hand off." This result provides a concrete improvement target rather than treating hand-off performance as complete.

2. Project Objectives and Scope

The project was structured around four practical objectives. First, build a chatbot from an admissions Knowledge Base. Second, make the chatbot ask a follow-up question when a user query is unclear. Third, test the assistant with normal, tricky, and out-of-scope questions. Fourth, measure accuracy and define a human hand-off rule.

- Knowledge-grounded answering: use the uploaded Knowledge Base as the source of truth.

- Clarification: ask one specific follow-up question when the user’s request is ambiguous and wait for the answer.
- No-assumption behavior: do not invent dates, fees, eligibility criteria, contacts, policies, or statistics.
- Human escalation: refer cases that require institutional judgment, unavailable information, privacy handling, complaints, or out-of-scope support.
- Evaluation: record expected and actual decisions and distinguish hand-off accuracy from hand-off rate.

3. System Design and Working

The prototype is intentionally lightweight. It is a documentation and evaluation package rather than a standalone deployed application. The core setup consists of a Gemini Gem, an uploaded Excel Knowledge Base, defined Gem instructions, a human hand-off policy, a manual test plan, and an evaluation sheet.

The Gem instructions establish the assistant’s behavior. The Knowledge Base is treated as the source of truth. For routine admissions questions—such as published deadlines, attendance requirements, or stated programme policies—the bot should provide a concise answer when the required information is available. When a question has multiple possible interpretations, the bot should ask one clarification question and stop rather than answer under an assumption.

The human hand-off policy is an important safety and reliability layer. It prevents the chatbot from presenting itself as an authority for individualized admission decisions, complaints, private information, or unsupported information. The project documentation also states that the bot must not claim that a live transfer or ticket was created unless the system actually supports that action.

4. Testing and Evaluation

The manual test plan covers factual answering, clarification behavior, unsupported claims, and hand-off decisions. The five completed hand-off cases provide the measurable evaluation included in the generated project package.

ID	Expected decision	Actual decision	Correct?	Notes
H1	Hand off	Hand off	Yes	Missing or private information
H2	Hand off	Hand off	Yes	Individual admission exception
H3	Hand off	Do not hand off	No	Complaint/dispute
H4	Do not hand off	Do not hand off	Yes	Routine FAQ
H5	Hand off	Hand off	Yes	Outside scope

Based on the completed sheet, hand-off decision accuracy is calculated as $4/5 \times 100 = 80\%$. The hand-off rate is $3/5 \times 100 = 60\%$. These metrics answer different questions: accuracy measures whether the bot chose the correct escalation decision, while hand-off rate measures how often escalation actually occurred. A high hand-off rate is not automatically desirable because unnecessary escalation can reduce the usefulness of an automated helpdesk.

5. Key Finding: Refund Complaint Handling

The most important finding is H3. The prompt states: “I am unhappy with my refund. I want to file a complaint and get my money back.” The expected decision was human hand-off because the case is a complaint/dispute. The recorded actual decision was “Do not hand off,” so this case was marked incorrect. This single case reduced the measured hand-off decision accuracy from a perfect score to 80%.

The result indicates that the hand-off rule needs to be reinforced specifically for complaints, disputes, and unresolved refund issues. The project already defines these situations as human-escalation cases, so the improvement should focus on making the operational rule more consistently recognized by the chatbot rather than expanding the scope of the bot.

6. Recommendations

- Strengthen complaint/dispute detection. Add explicit instruction examples for refund complaints, disputes, dissatisfaction, and requests for recovery of money. These should trigger human review.
- Use decision-oriented test cases. Repeat H3 with several differently worded complaints to determine whether the failure is wording-specific or reflects a broader classification problem.
- Separate routine FAQ handling from escalation handling. Routine questions such as deadlines should be answered directly, while exceptions, complaints, and individualized decisions should follow the escalation policy.
- Keep the Knowledge Base authoritative and maintained. Whenever fees, deadlines, policies, contacts, or programme rules change, update the Knowledge Base and rerun relevant tests.
- Maintain evidence of real testing. The project documentation recommends genuine screenshots of the Gem setup and test conversations; mockups should not be used as proof of testing.
- Track both accuracy and unnecessary escalation. A future evaluation should record false escalations as well as missed escalations, because optimizing only for a high hand-off rate can make the bot less useful.
- Protect privacy. Do not publish private student information, credentials, or confidential material in the repository or screenshots.
- Expand the test set after every instruction change. Changes to ambiguity handling or escalation rules can affect previously successful behaviors, so regression testing should be part of the workflow.

7. Reflection and Learning

This project demonstrates that building a useful chatbot is not only a matter of generating answers. The more important design challenge is defining when the system should answer, when it should ask for clarification, and when it should stop and involve a human. The project's no-assumption rule is particularly valuable because admissions questions can involve fees, eligibility, deadlines, refunds, and personal circumstances where an unsupported answer could mislead a student.

A second learning point is the importance of evaluation design. The project distinguishes functional FAQ testing from hand-off decision testing. This makes the results more meaningful: a chatbot can answer routine questions accurately while still needing improvement in escalation behavior. The H3 result illustrates this clearly. Instead of reporting only an overall impression of the chatbot, the evaluation identifies a specific failure mode that can be corrected and retested.

The clarification requirement also shows how conversational design affects reliability. For example, when a user asks for total cost including hostel without specifying the programme, the bot should first ask which programme the user means and wait. Providing a conditional answer at the same time could undermine the purpose of clarification. This demonstrates the value of precise instructions and controlled conversation flow.

Finally, the project provided practical exposure to AI-assisted workflow design: preparing a structured knowledge source, writing behavioral instructions, defining escalation criteria, creating a manual test plan, recording observed outcomes, and calculating evaluation metrics. The main lesson is that reliability comes from the combination of grounded information, clear rules, testing, and human oversight rather than from the chatbot model alone.

8. Conclusion

The College Admissions Helpdesk Bot successfully establishes a practical prototype for a knowledge-grounded admissions assistant. Its design covers routine FAQ answering, clarification of ambiguous requests, prevention of unsupported claims, and human hand-off. The completed hand-off evaluation achieved 80% decision accuracy and a 60% actual hand-off rate across five recorded cases. The H3 refund complaint is the principal issue identified in the completed evaluation and should be addressed before treating the hand-off workflow as fully reliable.

The next iteration should strengthen complaint/dispute recognition, expand the hand-off test set, rerun regression tests, and maintain evidence of genuine chatbot behavior. With these improvements, the prototype can provide a clearer demonstration of how a generative AI assistant can support admissions teams while preserving appropriate human oversight.